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Centro de Ciências Sociais

Instituto de Estudos Sociais e Políticos

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Dissertação apresentada, como requisito parcial para obtenção do título de Mestre, ao Programa de Pós-Graduação em Ciência Política, da Universidade do Estado do Rio de Janeiro.

Orientador: Prof. Dr. First name Last name

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DEDICATION

Dedication text.

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ABSTRACT

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RESUMO

LAST NAME, F. *Título da dissertação ou tese*. yyyy. 14 f. Dissertação (Mestrado em Ciência Política) – Instituto de Estudos Sociais e Políticos, Universidade do Estado do Rio de Janeiro, Rio de Janeiro, yyyy.

Texto do resumo em português. O resumo deve conter entre 150 e 500 palavras conforme as normas da UERJ.

Palavras-chave: Primeira palavra-chave. Segunda palavra-chave. Terceira palavra-chave.

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INTRODUCTION

To the worm that first gnawed the cold flesh of my corpse I dedicate these Posthumous Memoirs as a fond remembrance (Assis, 1881).

In statistics, “all models are wrong, but some are useful” (Box, 1976). A model is a deliberate simplification of reality, and the craft of modelling lies in choosing which aspects to keep. Gelman; Hill (2007) show how regression and hierarchical models negotiate this tension between parsimony and realism.

The quality of our conclusions depends on both the model and the data. A striking example is the forecasting of U.S. elections from a sample of Xbox users — heavily biased relative to the electorate — which, after suitable statistical adjustment, produced estimates competitive with traditional polls (Wang *et al.*, 2015).

1 ORDINARY LEAST SQUARES

Consider the linear regression model in matrix form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (1)$$

where $\mathbf{y}$ is the $n \times 1$ response vector, $\mathbf{X}$ is the $n \times p$ design matrix, $\boldsymbol{\beta}$ is the coefficient vector and $\boldsymbol{\varepsilon}$ is the error vector.

1.1 Closed-form derivation

The ordinary least squares estimator minimises the residual sum of squares:

$$S(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Expanding the product and using that $\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y}$ is a scalar (hence equal to its transpose),

$$S(\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}.$$

The first-order condition follows by differentiating with respect to $\boldsymbol{\beta}$ and setting the gradient to zero:

$$\frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} \stackrel{!}{=} \mathbf{0}.$$

Assuming $\mathbf{X}^\top \mathbf{X}$ is invertible, we solve for the estimator in its celebrated closed form:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (2)$$

1.2 Illustration

Figure 1 shows a set of points (x_i, y_i) and the line fitted by Equation 2. Table 1 summarises the estimates, with their standard errors and t -statistics.

Table 1 - Ordinary least squares estimates. Source: prepared by the author (2026).

Coefficient	Estimate	Std. error	<i>t</i> -statistic
β_0 (intercept)	1.81	0.31	5.80
β_1 (slope)	0.84	0.05	17.29

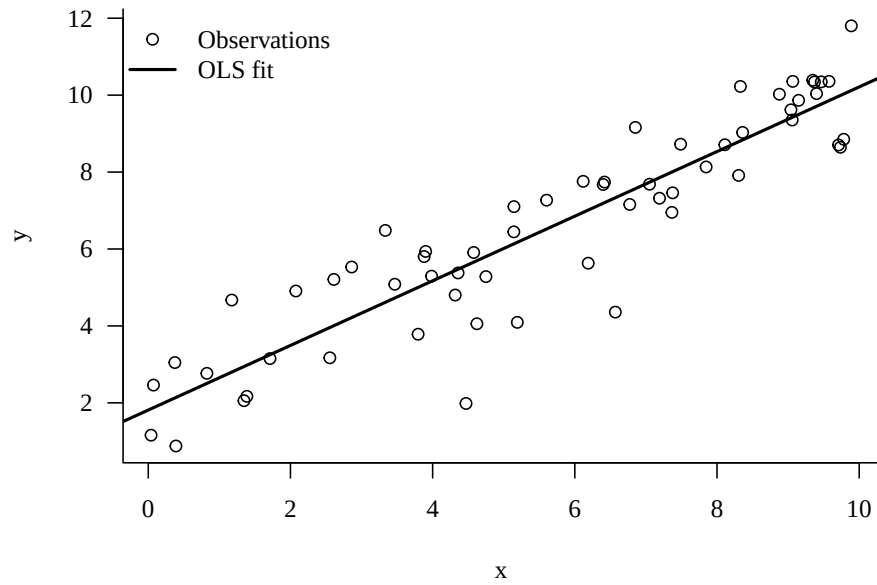


Figure 1 - Regression line fitted by ordinary least squares. Source: prepared by the author (2026).

CONCLUSION

Models are useful when they help us learn from data (Box, 1976), and the closed form of OLS in Equation 2 is the starting point for richer extensions, such as the hierarchical models of Gelman; Hill (2007).

REFERENCES

ASSIS, Machado de. **Memórias póstumas de brás cubas**. Rio de Janeiro: Tipographia Nacional, 1881.

BOX, George E. P. Science and statistics. **Journal of the American Statistical Association**, [n.p.], v. 71, n. 356, p. 791–799, 1976.

GELMAN, Andrew; HILL, Jennifer. **Data analysis using regression and multilevel/hierarchical models**. New York: Cambridge University Press, 2007.

WANG, Wei; ROTHCHILD, David; GOEL, Sharad; GELMAN, Andrew. Forecasting elections with non-representative polls. **International Journal of Forecasting**, [n.p.], v. 31, n. 3, p. 980–991, 2015.